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Hyperconvexity and domains of discontinuity for surface groups in $SL_3\mathbb{C}$

BY

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THESIS [PhD candidates may use DISSERTATION here instead]

Submitted as partial fulfillment of the requirements
for the degree of Doctorate of Philosophy in Mathematics
in the Graduate College of the
University of Illinois Chicago, 2026

Chicago, Illinois

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Dedication

In memory of my grandmother Martha, who shared her love of mathematics with me and many others.

Acknowledgements

TODO: Write acknowledgements

Preface

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Summary

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CHAPTER 1

Introduction

TODO: Write an introduction

CHAPTER 2

Background

TODO: Adding sections and background as necessary. TODO: Def Anosov TODO: More about flag varieties

1. Anosov representations in $SL_3\mathbb{C}$

Consider the collection of Anosov representations

$$\mathcal{A}_{\mathbb{R}}(\Gamma) := \{\rho : \Gamma \to SL_3\mathbb{R} \text{ Anosov}\}, \text{ and } \mathcal{A}(\Gamma) := \{\rho : \Gamma \to SL_3\mathbb{C} \text{ Anosov}\}. \tag{1}$$

Let $\rho_0 : \Gamma \to PSL_2\mathbb{R}$ Fuchsian and $\rho_i : \Gamma \to PSL_2\mathbb{C}$ quasi-Fuchsian. To create Anosov representations in $SL_3\mathbb{C}$, we can compose with an irreducible representation $\iota : PSL_2\mathbb{R} \to SL_3\mathbb{R}$:

$$\rho_F = \iota \circ \rho_0 : \Gamma \to SL_3\mathbb{R}, \text{ and } \rho_{QF} = \iota \circ \rho_i : \Gamma \to SL_3\mathbb{C}. \tag{2}$$

Similarly, (assuming that ρ_0, ρ_i lift to $SL_2\mathbb{R}$) we can compose with a reducible representation $\Delta : SL_2\mathbb{C} \to SL_3\mathbb{C}$:

$$\rho_R = \Delta \circ \rho_0 : \Gamma \to SL_3\mathbb{R}, \text{ and } \rho_{QR} = \Delta \circ \rho_i : \Gamma \to SL_3\mathbb{C}. \tag{3}$$

The representations ρ_{Fuchs} and ρ_{Red} are Anosov. We can then define the following subsets of $\mathcal{A}$:

- $SL_3\mathbb{C}$ quasi-Fuchsian representations $\mathcal{H}(\Gamma) := \{\rho \in \mathcal{A}(\Gamma) : \rho = \iota\circ$in the same Anosov component a
- Hitchin representations $\mathcal{H}(\Gamma) := \{\rho \in \mathcal{A}_{\mathbb{R}}(\Gamma) : \rho$in the same Anosov component as $\rho_0\}$
- Quasi-Hitchin representations
- $SL_3\mathbb{C}$-reducible representations
- Barbot representations
- Quasi-Barbot representations

2. The subvarieties of $\mathcal{F}^3$

2.1. Richardson varieties in $\mathcal{F}^3$. For any flag $x \in \mathcal{F}^3$, we define $Sch(x)$ to be the *principle Schubert varieties* centered at x. These are the closure of the principle Schubert cells, each identified by an element of the Weyl group $W(G) = A_2 = Sym(3)$. Let $x = (l, H) \in \mathcal{F}^3$ where $l \in \mathbb{CP}^2$ and $H \in (\mathbb{CP}^2)^*$ such that $l \subset H$. Then $Sch(x)$ consists of:

- $\alpha_{Id}(x) = \{(l, H)\} = \{x\}$,
- $\alpha_1(x) = \{(l, H_l) : l \subset H_l\}$,
- $\alpha_2(x) = \{(l_H, H) : l_H \subset H\}$,
- $\alpha_{12}(x) = \{(k, H_l) : l \subset H_l\}$,
- $\alpha_{21}(x) = \{(l_H, G) : l_H \subset H\}$, and
- $\alpha_{121}(x) = \alpha_{212}(x) = \mathcal{F}^3$.

We also define Sch as the *types* of principle Schubert varieties, e.g. α_1 or α_2, without specifying a basepoint x. For $g \in W(G)$, the complex dimension of $\alpha_g \in Sch$ is length of g in the Bruhat order. More explicitly, $dim(\alpha_{Id}) = 0$, $dim(\alpha_1) = dim(\alpha_2) = 1$, $dim(\alpha_{12}) = dim(\alpha_{21}) = 2$, and $dim(\alpha_{121}) = 3$.

We call a pair of flags $x, y \in \mathcal{F}^3$ *transverse* if they are in general position, meaning that for any proper, principle Schubert variety α, then $x \notin \alpha(y)$ and $y \notin \alpha(x)$.

For distinct flags $x, y \in \mathcal{F}^3$ such that either $l_x = l_y$ or $H_x = H_y$, then we define the *span* of x and y as follows:

$$span(x, y) := \begin{cases} \alpha_1(x) = \alpha_1(y) : l_x = l_y \\ \alpha_2(x) = \alpha_2(y) : H_x = H_y. \end{cases} \tag{4}$$

2.2. Richardson 0-varieties. *TODO: define Richardson variety more generally, then specialize to irreducible 0-varieties and irreducible 1-varieties.*

Let $x, y \in \mathcal{F}^3$ transverse. Define $R^0_{x,y}$ to be their irreducible *0-dimensional Richardson varieties*, which are the nonempty intersections between principle Schubert varieties $\alpha(x) \cap \beta(y)$ for $\alpha, \beta \in Sch$ such that $dim(\alpha) + dim(\beta) = dim(\mathcal{F}^3)$. There are exactly six such varieties $\{x, x_y, x^y, y_x, y^x, y\}$, where:

- $x := \alpha_{Id}(x) \cap \alpha_{121}(y)$,
- $x_y := \alpha_1(x) \cap \alpha_{12}(y)$,
- $x^y := \alpha_2(x) \cap \alpha_{21}(y)$,
- $y_x := \alpha_{12}(x) \cap \alpha_1(y)$,
- $y^x := \alpha_{21}(x) \cap \alpha_2(y)$, and
- $y := \alpha_{121}(x) \cap \alpha_{Id}(y)$.

Given two tranverse flags, the Richardson 0-varieties may be computed explicitly.

Example 2.1. *Choose x and y are the attracting and repelling flags of $\rho(g) = diag(\lambda, \mu, \nu)$ where $|\lambda| > |\mu| > |\nu|$. Then the six Richardson 0-varieties of x and y are $x = (e_1, e_1 + e_2)$, $x_y = (e_1, e_1 + e_3)$, $x^y = (e_2, e_1 + e_2)$, $y_x = (e_3, e_1 + e_3)$, $y^x = (e_2, e_2 + e_3)$, and $y = (e_3, e_2 + e_3)$.*

2.3. Richardson 1-varieties. Let $x, y \in \mathcal{F}^3$ transverse, and define $R^1_{x,y}$ to be their irreducible *1-dimensional Richardson varieties*. These are nonempty intersections between principle Schubert varieties $\alpha(x) \cap \beta(y)$ for $\alpha, \beta \in Sch$ such that $dim(\alpha) + dim(\beta) = dim(\mathcal{F}^3) + 1$. There are six such 1-varieties, each homeomorphic to $\mathbb{CP}^1$, which we distinguish using pairs of elements of $R^0_{x,y}$.

- $span(x, x_y) := \alpha_1(x) \cap \alpha_{121}(y)$,
- $span(x, x^y) := \alpha_2(x) \cap \alpha_{121}(y)$,
- $span(x_y, y_x) := \alpha_{12}(x) \cap \alpha_{12}(y)$,
- $span(x^y, y^x) := \alpha_{21}(x) \cap \alpha_{21}(y)$,
- $span(y_x, y) := \alpha_{121}(x) \cap \alpha_1(y)$, and
- $span(y^x, y) := \alpha_{121}(x) \cap \alpha_2(y)$.

The 1-varieties are not pairwise disjoint, e.g. $\{x\} = span(x, x_y) \cap span(x, x^y)$. The intersections of adjacent Richardson 1-varieties are exactly the Richardson 0-varieties. The connectivity of the 0- and 1-varieties is cyclic in the following way:

$$x \in span(x, x_y) \ni x_y \in span(x_y, y_x) \ni y_x \in span(y, y_x) \ni y \tag{5}$$

$$x \in span(x, x^y) \ni x^y \in span(x^y, y^x) \ni y^x \in span(y, y^x) \ni y. \tag{6}$$

Equivalently, in Figure 2.3 we describe the connectivity using a graph with six vertices representing 0-varieties and six edges representing 1-varieties. Two vertices are connected if the span of their 0-varieties is an irreducible 1-variety.

Remark 2.1. *If $x, y \in \mathcal{F}^3$ transverse and $\alpha, \beta \in Sch$ such that*

$$dim(\alpha) + dim(\beta) < dim(\mathcal{F}^3), \tag{7}$$

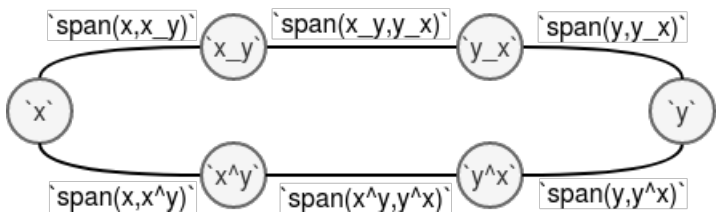

FIGURE 1. Graph representing the cyclic connectivity of Richardson 0- and 1-varieties for x and y transverse.

then $\alpha(x) \cap \beta(y) = \emptyset$.

3. Richardson varieties in $\mathcal{F}^n$ for $n \geq 4$

Conjectural picture for Richardson varieties in higher dimensions:

I suspect that topological conjugacies of Borel Anosov representations will preserve Richardson varieties up to $R^{n-2}_{x,y}$ organized into a cell decomposition with data an $(n-1)$-valent graph where vertices correspond to 0-varieties, edges correspond to 1-varieties, fully connected triples correspond to 2-varieties, etc. Matching/switching Richardson varieties will mean sending Richardson varieties to their dual.

I think balanced ideals I correspond to subgraphs of the Richardson variety graph containing exactly half of the vertices. In other words, $Th^I(x) \cap Th^I(y) = \emptyset$, and

$$(Th^I(x) \cap R^0_{x,y}) \cup (Th^I(y) \cap R^0_{x,y}) = R^0_{x,y}. \tag{8}$$

However, topological conjugacies should not depend on the choice of balanced ideal, so then they determine diffeomorphisms between all types of quotient manifolds. In other words, if $F : \mathcal{F}^n \to \mathcal{F}^n$ a topological conjugacy between ρ, ρ', then:

$$F(M^I_\rho) = M^I_{\rho'} \text{ for all balanced ideals } I \subset W(SL_n\mathbb{C}). \tag{9}$$

Most importantly, I believe the conclusions of Theorem 3.1 will apply in higher rank:

CONJECTURE 2.1. *For $\rho, \rho' : \Gamma \to SL_n\mathbb{C}$ Borel Anosov and topologically conjugate. Then ρ hyperconvex if and only if ρ' hyperconvex.*

CHAPTER 3

Hyperconvexity is closed under topological conjugacy

1. Assumptions and notation

Let Γ be the fundamental group of a closed hyperbolic surface, and let

$$\rho, \rho' : \Gamma \to SL_3\mathbb{C} \tag{10}$$

Anosov representations. We use the following notation related to ρ and ρ'.

- $\xi, \xi' : \partial\Gamma \to \mathcal{F}^3$ are the continuous, transverse, ρ- or ρ'- equivariant, dynamics-preserving boundary maps.
- $\Lambda, \Lambda' \subset \mathcal{F}^3$ are the limit curves, which are the images of ξ, ξ'.
- $Th(x) = \alpha_1(x) \cup \alpha_2(x)$ is the balanced ideal thickening of some $x \in \mathcal{F}^3$.
- $Th(\Lambda) = \cup_{x\in\Lambda} Th(x)$ is the thickening of the limit curve.

We call ρ, ρ' *topologically conjugate* if there exists a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that for all $g \in \Gamma$ and all $x \in \mathcal{F}^3$, then $F(\rho(g) \cdot x) = \rho'(g) \cdot F(x)$. We call F the *topological conjugacy* between ρ and ρ'.

Recall that ρ is *hyperconvex* if for all $x, y, z \in \Lambda^{(3)}$ then $l_z \not\subset l_x + l_y$ and $(H_x \cap H_y) \not\subset H_z$.

For $x, y \in \mathcal{F}^3$ transverse, recall the Richardson varieties of x and y, denoted by $R^0_{x,y}$ and $R^1_{x,y}$.

2. Hyperconvexity is closed under topological conjugacy

THEOREM 3.1. *If ρ and ρ' are topologically conjugate, then ρ is hyperconvex if and only if ρ' is hyperconvex.*

We use two supporting lemmas to prove this theorem. First, we show that the relative position of 0-dimensional Richardson varieties of Λ determines whether ρ is hyperconvex.

LEMMA 3.1.1. *An Anosov representation ρ is hyperconvex* if and only if *for all $x, y, z \in \Lambda^{(3)}$ then $R^0_{x,y} \cap R^0_{x,z} = \{x\}$.*

For the second lemma, we assume that $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'. Clearly $F(\Lambda) = \Lambda'$. This mapping is pointwise, i.e. for all $\alpha \in \partial\Gamma$, then $F(\xi(\alpha)) = \xi'(\alpha)$. So for $x \in \Lambda$, we can use the notation $x' := F(x) \in \Lambda'$.

We say F *preserves* the Richardson varieties of Λ if for all $x, y \in \Lambda^{(2)}$, then $F(R^0_{x,y}) = R^0_{x',y'}$ and $F(R^1_{x,y}) = R^1_{x',y'}$.

We will show that F preserves the Richardson varieties of Λ in one of two ways:

- We say F *matches* Richardson varieties if for all $x, y \in \Lambda^{(2)}$, then $F(x_y) = x'_{y'}$ and $F(span(x, x_y)) = span(x, x'_{y'})$.
- We say F *switches* Richardson varieties if for all $x, y \in \Lambda^{(2)}$, then $F(x_y) = x'^{y'}$ and $F(span(x, x_y)) = span(x, x'^{y'})$.

LEMMA 3.1.2. *If F is a topological conjugacy between ρ and ρ', then F preserves the Richardson varieties of Λ. Furthermore, F either matches or switches Richardson varieties.*

We also highlight the following consequence of the argument in Lemma 3.1.2:

COROLLARY 3.1.1. *For all $x \in \Lambda$, then $F(Th(x)) = Th(x')$.*

3. Proof of Theorem 3.1.

We first prove our main theorem using our supporting lemmas.

Proof. (Theorem 3.1) Suppose $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between Anosov representations $\rho, \rho' :\to SL_3\mathbb{C}$.

We begin with the forward direction. Assume that ρ is hyperconvex. Let $x, y, z \in \Lambda^{(3)}$. By Lemma 3.1.1, then $R^0_{x,y} \cap R^0_{x,z} = \{x\}$. By Lemma 3.1.2, then

$$F(R^0_{x,y}) = R^0_{x',y'}, \tag{11}$$

$$F(R^0_{x,z}) = R^0_{x',z'}. \tag{12}$$

Since F is a homeomorphism, then

$$R^0_{x',y'} \cap R^0_{x',z'} = F(R^0_{x,y} \cap R^0_{x,z}) \tag{13}$$

$$= F(x) \tag{14}$$

$$= \{x\}. \tag{15}$$

Since x, y, z were general, then this holds for all triples in $x', y', z' \in \Lambda'$, and ρ' is hyperconvex.

The other direction proceeds similarly. Assume that ρ is *not* hyperconvex. By Lemma 3.1.1, there exist $x, y, z \in \Lambda^{(3)}$ such that $R^0_{x,y} \cap R^0_{x,z} \neq \{x\}$. By Lemma 3.1.2, then

$$R^0_{x',y'} \cap R^0_{x',z'} = F(R^0_{x,y}) \cap F(R^0_{x,z}) \tag{16}$$

$$= F(R^0_{x,y} \cap F^(R^0_{x,z})) \qquad \neq F(x). \tag{17}$$

So in particular $R^0_{x',y'} \cap R^0_{x',z'} \neq \{x'\}$, so ρ' is *not* hyperconvex. □

We also prove our lemma that hyperconvexity is equivalent to the relative position of the Richardson 0-varieties of Λ.

Proof. (Lemma 3.1.1) Suppose ρ is Anosov. Let $x, y, z \in \Lambda^{(3)}$. Since x, y, z are pairwise transverse, then clearly

$$y_x, y^x, z_x, z^x \notin R^0_{x,y} \cap R^0_{x,z}. \tag{18}$$

To show that $R^0_{x,y} \cap R^0_{x,z} = \{x\}$, it suffices to check the pairs $x_y \neq x_z \in \alpha_1(x)$ and $x^y \neq x^z \in \alpha_2(x)$.

By definition, $x_y = (l_x, l_x + l_y)$ and $x_z = (l_x, l_x + l_z)$, which are distinct if and only if $l_x + l_y \neq l_x + l_z$, or equivalently if and only if $l_z \not\subset l_x + l_y$.

Similarly, $x^y = (H_x \cap H_y, H_x)$ and $x^z = (H_x \cap H_z, H_x)$, which are distinct if and only if $H_x \cap H_y \neq H_x \cap H_z$, or equivalently if and only if $H_x \cap H_y \not\subset H_z$.

So $R^0_{x,y} \cap R^0_{x,z} = \{x\}$ for all $x, y, z \in \Lambda^{(3)}$ is equivalent to ρ hyperconvex. □

The remainder of the chapter is devoted to the proof of Lemma 3.1.2, which says that a topological conjugacy F preserves irreducible Richardson varities of Λ.

4. Action of $\rho(g)$ on $\mathcal{F}^3$

In this section we describe the action of $\rho(g)$ on $\mathcal{F}^3$ for $g \in \Gamma$ infinite order. Then g has attracting and repelling group boundary points which we call $g^+, g^- \in \partial\Gamma$, respectively. Since ρ is Anosov, then $\rho(g)$ is strongly loxodromic, i.e. its eigenvalues have distinct magnitude. That means $\rho(g)$ has attracting and repelling limit curve points which we call $x := \xi(g^+)$ and $y := \xi(g^-)$, respectively.

Under the action $\rho(g)$, most of $\mathcal{F}^3$ is attracted to x and repelled by y. We define the *uphill* of x as the subset attracted to x under $\rho(g)$:

$$U_{\rho(g)}(x) := \{f \in F^3 : \rho(g^n) \cdot f \to x\}. \tag{19}$$

Similarly, we define the *downhill* of y as the subset repelled by y under $\rho(g)$, or equivalently the subset attracted by y under $\rho(g^{-1})$:

$$D_{\rho(g)}(y) = U_{rho(g^{-1})}(y) := \{f \in \mathcal{F}^3 : \rho(g^{-n}) \cdot f \to y\}. \tag{20}$$

We then define the *exceptional set* of $\rho(g)$ as the subset *not* attracted to x or *not* repelled by y under $\rho(g)$:

$$E_{\rho(g)} = \mathcal{F}^3 \setminus (U_{\rho(g)}(x) \cap D_{\rho(g)}(y)). \tag{21}$$

The set $E_{\rho(g)}$ is nonempty and closed, since it is complement of an open set. It contains the fixed points of $\rho(g) \cdot \mathcal{F}^3$, which are precisely $R^0_{x,y}$. It suffices to check this for $x, y, \rho(g)$ as in Example 2.1, since $\rho(g)$ is conjugate to $diag(\lambda, \mu, \nu)$.

We can partition $E_{\rho(g)} \setminus R^0_{x,y}$ into connected subsets using the attracting and repelling fixed points. By taking the closure of these subsets, we recover the 1-dimensional Richardson varieties $R^1_{x,y}$.

- $span(x, x_y) = \alpha_1(x) = \overline{U(x) \cap D(x_y)}$
- $span(x, x^y) = \alpha_2(x) = \overline{U(x) \cap D(x^y)}$
- $span(x_y, y_x) = \overline{U(x_y) \cap D(y_x)}$
- $span(x^y, y^x) = \overline{U(x^y) \cap D(y^x)}$
- $span(y_x, y) = \alpha_1(y) = \overline{U(y_x) \cap D(y)}$
- $span(y^x, y) = \alpha_2(y) = \overline{U(y^x) \cap D(y)}$

So $E_{\rho(g)}$ is exactly the union of the Richardson 1-varieties. The cyclic connectivity of the Richardson 0- and 1-varieties, and therefore that of the exceptional set, is described explicitly Equation 6 and Figure 2.3.

5. F preserves the Richardson varieties of pairs $\xi(g^+), \xi(g^-)$

Assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$. We use the notation $x' := F(x), y' := F(y)$.

First, we simplify the conclusions that we need to check.

Proposition 3.1. *Suppose that F a topological conjugacy such that F preserves the Richardson 1-varieties of a pair x and y in Λ. In other words, for some pair $x, y \in \Lambda^{(2)}$, we assume $F(R^1_{x,y}) = R^1_{x',y'}$. Then*

1. *F preserves the Richardson 0-varieties of x and y, i.e. $F(R^0_{x,y}) = R^0_{x',y'}$.*
2. *F either matches or switches the Richardson varieties of x and y.*

Proof. Let F be a homeomorphism that preserves the Richardson 1-varieties of x and y. First, let $z \in R^0_{x,y}$. Then $z = U \cap V$ for distinct $U, V \in R^1_{x,y}$. Since F is a homeomorphism, then

$$F(z) = F(U \cap V) \tag{22}$$

$$= F(U) \cap F(V). \tag{23}$$

By our assumption, then there exist $U', V' \in R^1_{x',y'}$ such that $U' = F(U)$ and $V' = F(V)$. Observe that $z' = U' \cap V'$ is therefore a Richardson 0-variety, so then $F(R^0_{x,y}) \subset F(R^0_{x',y'})$. Since F is a homeomorphism, then it restricts to a bijection on finite sets and $F(R^0_{x,y}) = R^0_{x',y'}$.

Recall that $x' = F(x), y' = F(y) \in \Lambda'$. Since $x \in span(x, x_y) \cap span(x, x^y)$, then $x' \in F(span(x, x_y)) \cap F(span(x, x^y))$, so either $F(span(x, x_y)) = span(x', x'_{y'})$ or $F(span(x, x_y)) = span(x, x^y)$. In the first case, F matches the Richardson varieties of x, y and $F(x_y) = x'_{y'}$, and in the second case F switches the Richardson varieties of x, y and $F(x_y) = x'^{y'}$. □

Next, we show that F preserves Richardson 1-varieties. We consider the case where x, y are the attracting and repelling flags of an infinite order element $g \in \Gamma$.

Proposition 3.2. *Let $x, y \in \Lambda$ such that $x = \xi(g^+)$ and $y = \xi(g^-)$. Then F preserves the Richardson 1-varieties of x and y.*

Proof. Since F is a topological conjugacy, it must preserve the exceptional set of $\rho(g)$, so $F(E_{\rho(g)}) = E_{\rho'(g)}$. Recall that the exceptional set of $\rho(g)$ is the union of the Richardson 1-varieties of x and y, that is:

$$E_{\rho(g)} = \cup_{V \in R^1_{x,y}} V. \tag{24}$$

Since $F|_{E_{\rho(g)}}$ is continuous and injective, then it must send 1-varieties to 1-varieties, so $F(R^1_{x,y}) = R^1_{x',y'}$. □

6. F preserves Richardson varieties of Λ

In this section, we extend our argument in Prop 3.2 from pairs $\xi(g^+), \xi(g^-)$ to all distinct pairs $\Lambda^{(2)}$. The main idea is that pairs of the form $\xi(g^+), \xi(g^-)$ are dense in Λ^2.

We use the following result from group theory:

Fact 3.1 (TODO REFERENCE). *Let Γ hyperbolic with boundary $\partial\Gamma \equiv S^1$, and let $x, z \in \partial\Gamma$. Then there exists a quasigeodesic sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that for all $y \in \partial\Gamma$:*

$$lim_{i\to\infty} g_i \cdot y = \begin{cases} x : y = x \\ z : y \neq x \end{cases} . \tag{25, 26}$$

We also observe that Richardson 1-varieties vary continuously on transverse pairs of flags:

Remark 3.1. *Let $\{x_i\}_{i=1}, \{y_i\}_{i=1} \in (\mathcal{F}^3)^{\mathbb{N}}$ such that for all $i \in \mathbb{N}$, the pair (x_i, y_i) are transverse. Furthermore, assume $x_i \to x$ and $y_i \to y$, and that the pair (x, y) is transverse. Then $lim_{i\to\infty} R^1_{x_i,y_i} = R^1_{x,y}$.*

The continuinity of Richardson 1-varieties also respects the type, e.g. $lim_{i\to\infty} span(x_i, x_i^{y_i}) = span(x, x^y)$.

So then we consider the Richardson varieties of pairs x, z where $x = \xi(g^+)$ and z is a (distinct) generic point on Λ.

Proposition 3.3. *Let $g \in \Gamma$ infinite order, and define $x := \xi(g^+)$. Let $z \in \Lambda \setminus \{x\}$. Then F preserves the Richardson 1-varieties of x and z.*

PROOF. (Proposition 3.3) If $z = \xi(g^-)$ then we are done by Prop 3.2, so assume not. Set $y := \xi(g^-)$.

Let $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ as in Fact 3.1, such that $\lim_{i=1} \rho(g_i) \cdot y = z$ and $lim_{i=1}\rho(g_i) \cdot x = x$. For all $i \in \mathbb{N}$, define the following:

$$x_i := \rho(g_i) \cdot x, \tag{27}$$

$$y_i := \rho(g_i) \cdot y, \tag{28}$$

$$x_i' := F(x_i), \text{ and} \tag{29}$$

$$y_i' := F(y_i). \tag{30}$$

By Prop. 3.2, then for all $i \in \mathbb{N}$, F preserves the Richardson 1-varieties of x_i and y_i. By Rem. 3.1, then $\lim_{i\to\infty} R^1_{x_i,y_i} = R^1_{x,z}$. By continuity of F, then $\lim_{i\to\infty} x_i' = x'$ and $\lim_{i\to\infty} y_i' = z'$. We apply Rem. 3.1 again to the transverse sequence (x_i', y_i'), so then

$$F(R^1_{x,z}) = \lim_{i\to\infty} F(R^1_{x_i,y_i}) \tag{31}$$

$$= \lim_{i\to\infty} R^1_{x_i',y_i'} \tag{32}$$

$$= R^1_{x',z'}. \tag{33}$$

□

Next, we extend Proposition 3.3 to Richardson varieties of pairs w, z where both w and z are distinct generic points of Λ.

PROPOSITION 3.4. *Let $w, z \in \Lambda^{(2)}$. Then F preserves the Richardson 1-varieties of w and z.*

PROOF. If w or z is the attracting fixed point of an infinite order element $g \in \Gamma$, then we are done by Prop 3.3. Assume not.

Let $\{x_i\}_{i=1} \in \Lambda^{\mathbb{N}}$ such that each $x_i = \xi(g_i)$ is the attracting fixed point of an infinite order word, and $\lim_{i\to\infty} x_i = w$. This sequence exists because attracting fixed points are dense in Λ. Define $x_i' = F(x_i)$ for all $i \in \mathbb{N}$. By continuity of F, then $\lim_{i\to\infty} x_i' = w'$.

The argument proceeds similarly to that of Prop. 3.3. By Prop. 3.3, then for all $i \in \mathbb{N}$, F preserves the Richardson 1-varieties of x_i and z. By Rem. 3.1, then $\lim_{i\to\infty} R^1_{x_i,z} = R^1_{w,z}$. We apply Prop. 3.1 again to the transverse sequence $(x_i', z') \to (w', z')$, so then:

$$F(R^1_{w,z}) = \lim_{i\to\infty} F(R^1_{x_i,z}) \tag{34}$$

$$= \lim_{i\to\infty} R^1_{x_i',z'} \tag{35}$$

$$= R^1_{w',z'}. \tag{36}$$

□

7. F matches or switches Richardson varieties

In this section, we show that F either matches all Richardson varieties of Λ or switches all Richardson varieties of Λ, which concludes the proof of Lemma 3.1.2. We also highlight that F preserves the balanced ideal thickening of each $x \in \Lambda$.

Recall that $Th(x) = \alpha_1(x) \cup \alpha_2(x)$, so $Th(x) = span(x, x_y) \cup span(x, x^y)$ for any $y \in \Lambda$ distinct from x.

Proof. (Corollary 3.1.1) Let $x \in \Lambda$, and let $y \in \Lambda \setminus \{x\}$. By Prop. 3.4, F preserves the Richardson 1-varieties of x and y. By Prop. 3.1, F either matches or switches the Richardson varieties of each pair x and y. So then

$$\begin{aligned} F(Th(x)) &= F(span(x, x_y) \cup span(x, x^y)) && (37)\\ &= F(span(x, x_y)) \cup F(span(x, x^y)) && (38)\\ &= span(x', x'_{y'}) \cup span(x', x'^{y'}) && (39)\\ &= Th(x'). && (40) \end{aligned}$$

□

If F matches (resp. switches) the Richardson varieties of some pair of $x, y \in \Lambda^{(2)}$, then F matches (resp. switches) the Richardson varieties of all such pairs.

Proof. Observe that

$$\begin{aligned} \alpha_1(\Lambda) &= \cup_{x \in \Lambda} \alpha_1(x) && (41)\\ &= \cup_{x,y \in \Lambda^{(2)}} span(x, x_y) && (42)\\ &\cong S^1 \times \mathbb{CP}^1 \subset \mathcal{F}^3. && (43) \end{aligned}$$

Similarly, $\alpha_2(\Lambda) \cong S^1 \times \mathbb{CP}^1$. In particular, $\alpha_1(\Lambda)$ and $\alpha_2(\Lambda)$ are connected subsets of $\mathcal{F}^3$ whose intersection is Λ.

By Cor. 3.1.1, clearly

$$\begin{aligned} F(\alpha_1(\Lambda)) \cup F(\alpha_2(\Lambda)) &= F(\alpha_1(\Lambda) \cup \alpha_2(\Lambda)) && (44)\\ &= F(Th(\Lambda)) && (45)\\ &= Th(\Lambda') && (46)\\ &= \alpha_1(\Lambda') \cup \alpha_2(\Lambda'). && (47) \end{aligned}$$

Since F is a homeomorphism which sends Λ to Λ', then either $F(\alpha_1(\Lambda)) = \alpha_1(\Lambda')$, or $F(\alpha_1(\Lambda)) = \alpha_2(\Lambda)$.

In other words, if F matches the Richardson varieties of one pair in $\Lambda^{(2)}$, then it matches the Richardson varieties of all such pairs. Otherwise, F switches the Richardson varieties of all such pairs. □

This completes the proof of Lemma 3.1.2.

8. Random remarks

This might belong better in a different chapter.

Remark 3.2. *For distinct $x, y \in \mathcal{F}^3 \setminus \Lambda$, F will not necessarily preserve the 0-dimensional Richardson varieties.*

Remark 3.3. *Theorem 3.1 applies to virtual hyperbolic surface groups as well, e.g. $\Gamma = \pi_1(\mathcal{O}^2)$ where $\mathcal{O}^2$ is a hyperbolic 2-orbifold.*

CHAPTER 4

Anosov components have one topological conjugacy class

1. Assumptions and notation

TODO assumptions on ρ, ρ'.

Recall the set of Anosov representations $\mathcal{A}(\Gamma)$, the quasi-Hitchin representations $\mathcal{QH}(\Gamma)$ and the quasi-Barbot representations $\mathcal{QB}(\Gamma)$.

Recall irreducible representations.

2. Anosov components have one topological conjugacy class

THEOREM 4.1. *Suppose $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ are irreducible representations in the same Anosov component. Then ρ, ρ' are topologically conjugate.*

Combined with Theorem 3.1, we get the following corollaries:

COROLLARY 4.1.1. *All quasi-Hitchin representations are hyperconvex. All quasi-Barbot representations are* not *hyperconvex.*

Furthermore:

COROLLARY 4.1.2. *The quasi-Hitchin and quasi-Barbot Anosov components are disjoint.*

COROLLARY 4.1.3. *The collection of hyperconvex representations is open and closed in $\mathcal{A}(\Gamma)$.*

We observe that distinct Anosov components may share a single topological conjugacy class. For example, the quasi-Barbot components in $Hom(T(p,q,r), SL_3\mathbb{C})$ are disjoint. However, they are dual, so have the same conjugacy class:

PROPOSITION 4.1. *Let $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov and let ρ^* its dual. Then ρ, ρ^* are topologically conjugate.*

We also describe the topological conjugacy classes of the connected components of reducible Anosov representations $\mathcal{A}_{red}(\Gamma) \subsetneq QB(\Gamma)$.

THEOREM 4.2. *There are exactly three topological conjugacy classes of $\mathcal{A}_{red}(\Gamma)$:*

- *The class of semisimple representations,*
- *The class of line- or plane-reducible representations, and*
- *The class of reducible, but neither line- nor plane-reducible representations.*

Representations in the last class are topologically conjugate to the irreducible representations in the same component of $QB(\Gamma)$.

3. Key ingredients for proof

In this section, we highlight prior results that are used in the proof of Theorem 4.1. First, we use the following theorem of Stecker:

THEOREM 4.3 (**??**, Thm 3.24). *Let $\rho : \Gamma \to G$ Borel Anosov. If $dim(\Lambda) \leq mbic(G) - 2$, then every non-empty cocompact domain of discontinuity in $\mathcal{F}^3$ is dense and connected, and there is a bijection:*

$$\{\textit{balanced ideals in } W(G)\} \leftrightarrow \{\textit{nonempty cocompact domains of discontinuity in } \mathcal{F}(G)\} \tag{48}$$

We tailor Stecker's theorem to our assumptions. Recall that $G = SL_3\mathbb{C}$, so $mbic(G) = 4$ and $\mathcal{F}(G) = \mathcal{F}^3$. Then $W(G) = A_2$ which has exactly one balanced ideal. In $SL_3\mathbb{C}$, all Anosov representations are Borel Anosov. Furthermore, we assume that Γ is a hyperbolic surface group, so $\partial\Gamma \cong S^1$. Then $1 \leq dim(\Lambda_\rho) \leq 2$, because ξ_ρ is an injective and continuous function of S^1. So the condition in Theorem 4.3 is met, and:

Remark 4.1. *For Γ a hyperbolic surface group and $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov, then ρ has a* unique *nonempty cocompact domain of discontinuity $\Omega_\rho \subset \mathcal{F}^3$. Furthermore, $\Omega_r ho$ is connected and dense in $\mathcal{F}^3$.*

We also use the following lemma, proved in the following section:

Lemma 4.3.1. *For Γ a hyperbolic surface group and $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov, then Ω_ρ is simply connected.*

Finally, we use the following theorem of Dumas and Sanders:

Theorem 4.4 (**??**, Thm 5.1). *Let (P, Q) be parabolic subgroups of G, with P symmetric, and let $I \subset W(G)$ be a balanced ideal of type (P, Q). Let $\rho : \Gamma \to G$ be a P-Anosov representation. Then for any $\rho' \in \mathcal{A}(\rho, P)$, the quotient manifolds M^I_ρ and $M^I_{\rho'}$ are diffeomorphic.*

We again tailor this theorem to our assumptions that $G = SL_3\mathbb{C}$, where there is only symmetric parabolic subgroup and one balanced ideal in $W(G)$. Then:

Remark 4.2. *For Γ a hyperbolic surface group and $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ in the same Anosov component, then*

$$M_\rho \equiv_{diffeo} M_{\rho'}. \tag{49}$$

4. Proof of Theorem 4.1

Proof. (Theorem 4.1) Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ irreducible and in the same Anosov component. We want to show the existence of $F : \mathcal{F}^3 \to \mathcal{F}^3$ a topological connjugacy of ρ, ρ'.

Then by Thm. 4.4, there exists a diffeomorphism $D : M_\rho \to M_{\rho'}$. This lifts to a diffeomorphism of the universal cover $\tilde{D} : \tilde{M}_\rho \to \tilde{M}_{\rho'}$. Then by Lemma 4.3.1, the domains of discontinuity Ω_ρ, $\Omega_{\rho'}$ are simply connected, so $\tilde{M}_\rho \equiv \Omega_\rho$ and $\tilde{M}_{\rho'} \equiv \Omega_{\rho'}$. So we can consider $\tilde{D} : \Omega_\rho \to \Omega_{\rho'}$.

Clearly $\tilde{D}$ is a topological conjugacy of Ω_ρ and $\Omega_{\rho'}$, since it is a diffeomorphism and for all $f \in \Omega_\rho$ and $g \in \Gamma$ we have $\tilde{D}(\rho(g) \cdot f) = \rho'(g) \cdot \tilde{D}(f)$.

Next, we want to extend $\tilde{D}$ to a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that $F|_{\Omega_\rho} \equiv \tilde{D}$. By Theorem 4.3, Ω_ρ is an open, connected dense subset of $\mathcal{F}^3$, so $\tilde{D}$ has a unique continuous extension to $\mathcal{F}^3$ using sequences in Ω_ρ which limit to points in $Th(\Gamma)$.

TODO check thet F is a topological conjugacy. TODO where do we use irreducible? Probably in the checking F top conj □

Remark 4.3. *The topological conjugacy F may not be unique, however it is uniquely determined by the choice of the diffeomorphism D and its lift $\tilde{D}$.*

Proof. (Lemma 4.3.1) Fact: $\pi_1\mathcal{F}^3$ is trivial. Fact: The odd homology groups $H^{2n+1}(\Omega_\rho)$ vanish. Fact: Then $\pi_1(\Omega_\rho)$ is perfect, since $\pi_1(\Omega_\rho)^{ab} = H^1(\Omega_\rho) = \{Id\}$.

Claim: First consider $\Omega_{\rho_{Fuchs}}$ where ρ_F is $SL_3\mathbb{C}$-Fuchsian. Then $\Omega_{\rho_{Fuchs}}$ is simply connnected.

Claim: If there is a transverse homotopy between $\Lambda_{\rho_{Fuchs}}$ and Λ_ρ, then $\pi_1(\Omega_{\rho_{Fuchs}}) = \pi_1(\Omega_\rho)$.

Claim: There is a transverse homotopy between $\Lambda_{\rho_{Fuchs}}$ and Λ_ρ. □

5. Topological conjugacies of reducible representations

This argument for Theorem 4.2 hinges on whether ρ fixes a line and/or a plane in $\mathbb{C}^3$. Let $\rho : \Gamma \to SL_3\mathbb{C}$ reducible. We say that ρ is

(1) *semisimple* if ρ fixes a line $l \in CP^2$ and a plane $H \in (CP^2)^*$,
(2) *line-reducible* if ρ fixes a line $l \in CP^2$, but no plane,
(3) *plane-reducible* if ρ fixes a plane $H \in (CP^2)^*$, but no line, and
(4) *neither line- nor plane-reducible* otherwise.

Given $l \in \mathbb{CP}^2$, we use the abuse of notation

$$\alpha_1(l) := \alpha_1(l, H) \tag{50}$$

for any $H \supset l$ because $\alpha_1(l, H)$ does not depend on choice of H. Similarly, for $H \in (\mathbb{CP}^2)^*$, we use the analogous abuse of notation

$$\alpha_2(H) := \alpha_2(l, H) \text{ for some } l \subset H. \tag{51}$$

If ρ is semisimple, then for all $x, y, z \in \Lambda_\rho^{(3)}$ we have $x_y = x_z$, $y_x = y_z$, $z_x = z_y$, and $x^y = x^z$, $y^x = y^z$, $z^x = z^y$. So ρ fixes the line $l = l_{x^y}$ and the plane $H = H_{x_y}$. This means that the following sets are invariant under ρ:

$$\alpha_1(l) = span(x_y, y_x) = span(x_z, z_x) = span(y_z, z_y), \text{ and} \tag{52}$$

$$\alpha_2(H) = span(x^y, y^x) = span(x^z, z^x) = span(y^z, z^y). \tag{53}$$

Similarly, if ρ is line-reducible (but not semisimmple), then all $x^y = x^z$, $y^x = y^z$, $z^x = z^y$ but not all $x_y = x_z$. So ρ also fixes the line $l = l_{x^y}$, so the following set is invariant under ρ:

$$\alpha_1(l) = span(x^y, y^x) = span(x^z, z^x) = span(y_z, z_y). \tag{54}$$

If ρ is plane-reducible (but not semisimple), then for all $x, y, z \in \Lambda_\rho^{(3)}$ we have $x_y = x_z$, $y_x = y_z$, $z_y = z_x$, but not all $x^y = x^z$. So ρ fixes the plane $H = H_{x_y}$, so the following set is invariant under ρ:

$$\alpha_2(H) = span(x_y, y_x) = span(x_z, z_x) = span(y_z, z_y). \tag{55}$$

Finally, if ρ is neither line- nor plane-reducible, then ρ fixes no complex line or plane in $\mathbb{C}^3$.

Proof. (Theorem 4.2) If ρ, ρ' are in the same representation conjugacy class and have the same reducible type, then they are topologically conjugate.

Now we consider pairs $\rho, \rho' \in \overline{[\rho]}$ with different reducible types.

Semisimple representations are only topologically conjugate to other semisimple representations.

Since line-reducible and plane-reducible representations but fix a complex line in Ω_ρ, then they can be topologically conjugate.

Finally, if ρ is neither line- nor plane-reducible, then it does not fix any complex line in Ω_ρ, and is therefore not topologically conjugate to other reducible types. However, it is topologically conjugate to other irreducible representations in the same Anosov component. □

We also gain the following corollary, which is immediate from the proof of the Theorem and Lemma 3.1.2:

COROLLARY 4.4.1. *Suppose F a topological conjugacy between ρ line-reducible and some reducible ρ'. If ρ' line-irreducible, then F matches Richardson varieties. Otherwise ρ' is plane-irreducible and F switches Richardson varieties.*

CHAPTER 5

Vitae

TODO: write a vitae

APPENDIX A

Technical Lemmas

TODO: Move any annoyingly technical lemmas here.

Bibliography

[1] E. Dumas, and A. Sanders "Geometry of compact complex manifolds associated to generalized quasi-Fuchsian varieties" (2017)
[2] F. Stecker, "Balanced ideals and domains of discontinuity of Anosov representations" (2022)